RWANDA TVET BOARD
COMPETENCY-BASED ASSESSMENT
[SUMMATIVE ASSESSMENT]

Program: SOFTWARE DEVELOPMENT | **Level:** 5 | **Total Marks:** 100
Module: Machine Learning Application | **Date:** 2026-05-21 | **Time:** 3 Hours

Name: ____________________________ Index No: ______________ Class: ________

SECTION A: TRUE / FALSE

Answer True (T) or False (F).

1. Chai directly affects the security of applying Data Pre-processing. **(3 marks)**

   Answer: **True** **False**

2. It is possible to complete developing Machine Learning Model without using PUT. **(3 marks)**

   Answer: **True** **False**

3. Response directly affects the security of developing Machine Learning Model. **(3 marks)**

   Answer: **True** **False**

4. Understanding REST API is required before Perform Model Deployment. **(3 marks)**

   Answer: **True** **False**

SECTION B: MULTIPLE CHOICE

Choose ONE correct answer and circle the letter.

5. How would you correctly apply GET when applying Data Pre-processing? **(3 marks)**

   A. List
   B. GET
   C. JWT
   D. Feature
   Answer: _______

6. What is the main purpose of PUT when applying Data Pre-processing? **(3 marks)**

   A. Mapping
   B. PUT
   C. Python
   D. Deployment
   Answer: _______

7. What is the key difference when using Schema for applying Data Pre-processing? **(3 marks)**

   A. Model
   B. Schema
   C. HTTPS
   D. ref
   Answer: _______

8. What is the key difference when using Model for developing Machine Learning Model? **(3 marks)**

   A. Object

B. JWT
C. Encryption
D. Model
Answer: _______

9. Which approach to GET is most effective when developing Machine Learning Model? **(3 marks)**

A. Data Types
B. Python
C. Express
D. GET
Answer: _______

10. Which design using RESTful API would best support Perform Model Deployment? **(3 marks)**

A. Model
B. Object
C. RESTful API
D. Encryption
Answer: _______

11. What is the key difference when using Authentication for Perform Model Deployment? **(3 marks)**

A. DELETE
B. Express
C. Authentication
D. Unsupervised Learning
Answer: _______

SECTION C: MATCHING

Match each term in Column A with Column B.

12. Match each term in Column A with its correct description in Column B. **(5 marks)**

Column A (Terms)	Column B (Descriptions)
1. Machine learning life cycle	A. is a cyclic process to build an efficient machine learning project.
2. Data Gathering	B. is the first step of the machine learning life cycle. The goal of this step is to
3. Data preparation	C. is a step where we put our data into a suitable place and prepare it to
4. Data wrangling	D. is the process of cleaning and converting raw data into a useable format. It

Answers: 1.___ 2.___ 3.___ 4.___

13. Match each term in Column A with its correct description in Column B. **(5 marks)**

Column A (Terms)	Column B (Descriptions)
1. Model deployment in machine learning	A. is the process of integrating a trained machine
2. Model deployment	B. is the process of integrating a trained machine learning
3. A Dataset	C. is a set of data grouped into a collection with which developers can work to
4. This	D. is the Iris dataset. Since this is a dataset with which we build models, there are

Answers: 1.___ 2.___ 3.___ 4.___

SECTION D: OPEN-ENDED

Answer ALL questions. Write in the spaces provided.

14. Demonstrate how to use Deployment when applying Data Pre-processing. Provide a step-by-step explanation. **(3 marks)**

15. Explain how HTTPS works in the context of applying Data Pre-processing. **(9 marks)**

16. Define Express and explain its role in developing Machine Learning Model. **(9 marks)**

17. Demonstrate how to use GET when developing Machine Learning Model. Provide a step-by-step explanation. **(9 marks)**

18. Evaluate the effectiveness of Express when developing Machine Learning Model. Justify with examples. **(9 marks)**

19. Analyze the importance of JWT in Perform Model Deployment. Identify advantages and limitations. **(9 marks)**

20. Demonstrate how to use Encryption when Perform Model Deployment. Provide a step-by-step explanation. **(9 marks)**

END OF EXAMINATION | Summative | Machine Learning Application | Total: 100 marks